

citemsp: Citation Locators for L^AT_EX

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Abstract

The `citemsp` package provides a single user-facing command, `\citemsp`, that attaches section (§), paragraph (§¶), and prefix-based locators directly to numeric citation labels. It works on top of either `biblatex` or `natbib`, auto-detecting which backend is loaded. An extensible prefix registry currently provides ten built-in types, and users can register custom prefixes with `\citemspprefix`. The package parses a slash-delimited syntax, rendering each locator as a superscript/subscript pair on the corresponding citation number, producing output such as [$1_{45}^{§3.2}$, $2_{500}^{§p.520}$, $3_{eq.4.3}^{§1}$]. Plain entries (no slashes) are batched and passed to the backend’s native `\cite`, so range compression (`numeric-comp`, `sort&compress`) works naturally¹.

1 Requirements

- L^AT_EX2e (2020/10/01 or later)
- `biblatex` (any numeric style) **or** `natbib` (with the `numbers` option)
- `graphicx`
- `etoolbox`
- `amssymb` — recommended for best glyph quality ($\trianglelefteq$, $\boxplus$). If not loaded, `citemsp` provides primitive fallback definitions.

2 Introduction

The `\citemsp` command accepts a comma-separated list of entries. Each entry has the form `key`, `key/loc1`, or `key/loc1/loc2`. By default, the first slot renders as § (section) and the second as §¶ (paragraph). A single-letter prefix changes the locator type:

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¹The package is available at <https://github.com/ApostolosTsabodimos/citemsp-Latex-package> and on CTAN.

<code>\citemsp{key}</code>	→	$[1]$
<code>\citemsp{key/section}</code>	→	$[1]^{§6.2}$
<code>\citemsp{key/section/paragraph}</code>	→	$[1]^{§6.2}_{¶3}$
<code>\citemsp{key//paragraph}</code>	→	$[1]_{¶3}$
<code>\citemsp{key/section/e4.3}</code>	→	$[1]^{§6.2}_{eq.4.3}$

Note the subscript-only form `key//par`: an empty first slot with a double slash gives a paragraph (or prefixed) subscript without a section superscript.

The current ten built-in prefixes are listed in Table 1. Each prefix can appear in either the superscript or subscript slot. Users can register additional prefixes with `\citemspprefix{letter}{label}`.

Prefix	Locator	Render	In super slot	In sub slot	Both slots
A	Appendix	App.	$[1]^{App.2}$	$[1]_{App.2}$	$[1]^{§3.2}_{App.2}$
C	Corollary	Cor.	$[1]^{Cor.3}$	$[1]_{Cor.3}$	$[1]^{§4.1}_{Cor.3}$
L	Lemma	Lem.	$[1]^{Lem.1}$	$[1]_{Lem.1}$	$[1]^{§3.2}_{Lem.1}$
T	Theorem	Th.	$[1]^{Th.4.2}$	$[1]_{Th.4.2}$	$[1]^{§6.1}_{Th.4.2}$
d	Definition	$\triangleq$	$[1]^{d.5}$	$[1]_{d.5}$	$[1]^{§3.2}_{d.5}$
e	Equation	eq.	$[1]^{eq.4.3}$	$[1]_{eq.4.3}$	$[1]^{§6.2}_{eq.4.3}$
f	Figure	fig.	$[1]^{fig.3}$	$[1]_{fig.3}$	$[1]^{§6.1}_{fig.3}$
n	Footnote	†	$[3]^{†7}$	$[3]_{†7}$	$[3]^{§1}_{†7}$
p	Page	pp.	$[1]^{pp.42}$	$[1]_{pp.42}$	$[1]^{§3.2}_{pp.42}$
t	Table	$\boxplus$	$[2]^{t.2}$	$[2]_{t.2}$	$[2]^{§8.4}_{t.2}$

Table 1: Built-in prefix registry.

Users can register custom prefixes with `\citemspprefix`. Place the command in the preamble (after `\usepackage{citemsp}`) so it is available throughout the document:

```
\usepackage{citemsp}
\citemspprefix{R}{Rem.}      "R" renders as "Rem."
\citemspprefix{w}{w.}       "w" renders as "w."
```

For example, after registering `R` for “Remark”: `\citemsp{cox1979/12.4/R3}` → $[4]^{§12.4}_{Rem.3}$, and after registering `w` for “widget”: `\citemsp{zongker2006/3/w5}` → $[5]^{§3}_{w5}$. The prefix letter must not collide with an existing built-in prefix.

3 Loading and Configuration

With biblatex

```
\usepackage[style=numeric-comp, backend=biber]{biblatex}
\addbibresource{refs.bib}
\usepackage{citemsp}
```

Use `numeric-comp` (or `numeric`) as the style. Range compression of consecutive plain entries requires `numeric-comp`.

Configuration

Two user-configurable parameters control the locator appearance:

- `\citemsscale` (default 0.35) — controls the locator glyph size via `\scalebox` from `graphicx`.
- `\citemspraiseoffset` (default 1.5pt) — vertical offset applied via `\raisebox` to superscript locators.

Redefine them after loading:

```
\renewcommand{\citemsscale}{0.4}
\renewcommand{\citemspraiseoffset}{2pt}
```

4 Examples

4.1 Single source with section and paragraph

The lack of randomised controlled trials on parachutes~\citemsp{smith2003/6.1/2} remains a gap in evidence-based medicine.

Output: The lack of randomised controlled trials on parachutes [1^{§6.1}_{¶2}] remains a gap in evidence-based medicine.

4.2 Multiple sources with mixed locators

Writer’s block~\citemsp{upper1974, smith2003/4, meyerrochow2003/1/2} has been studied from behavioural, medical, and ornithological perspectives.

Output: Writer’s block [6, 1^{§4}, 3^{§1}_{¶2}] has been studied from behavioural, medical, and ornithological perspectives.

4.3 Range compression

Plain entries (no slashes) are batched and passed to the backend’s native `\cite`, so range compression works automatically when using `numeric-comp` (biblatex) or `sort&compress` (natbib). Entries *with* locators are rendered individually and interleaved with the compressed group.

1. All plain

```
\citemsp{upper1974, pennycook2015, smith2003, meyerrochow2003, wang2020}
```

Output: [1–3, 6, 7]

2. Plain entries mixed with locator entries:

```
\citemsp{smith2003/4.1, upper1974, pennycook2015, meyerrochow2003, wang2020}
```

Output: [1^{§4.1}, 2, 3, 6, 7]

3. Mixed entry locators

```
\citemsp{upper1974/1, smith2003/4.1/2,meyerrochow2003/e14, wang2020/p21}
```

Output: [6^{§1}, 1^{§4.1}₂, 3^{eq.14}, 2^{pp.21}]

```
See~\citemsp{smith2003/6.1/2, matthews1995/e6.55, wang2020/8.4/t2,
yap2022/p170} for parachutes, toast, graphene, and dead spiders respectively.
```

Output: See [1^{§6.1}₂, 8^{eq.6.55}, 2^{§8.4}₂, 9^{pp.170}] for parachutes, toast, graphene, and dead spiders respectively.

4.4 Font contexts

The package works correctly in all standard L^AT_EX font contexts. By default, citations inherit the surrounding font shape and weight. For italic and slanted shapes, a small kern is automatically inserted to prevent the locator from colliding with the slanted closing bracket.

Default behaviour (citations inherit surrounding font):

Upright: [1^{§6.1}₂]
Italic: [*1^{§6.1}*₂]
Slanted: [*1^{§6.1}*₂]
Bold: [**1^{§6.1}**₂]
Bold italic: [***1^{§6.1}***₂]
SMALL CAPS: [1^{§6.1}₂]
Emph (italic): [*1^{§6.1}*₂] — nested emph (upright): [1^{§6.1}₂]

To force citations into upright, medium-weight, default-family text regardless of context, enable fixed-font mode with `\citemspfixedfonttrue` in the preamble. When active, the entire `\citemsp` output is wrapped in `\textnormal`, so an example *italic* [1^{§6.1}₂] and **bold** [1^{§6.1}₂] contexts both produce the same upright citation.

5 Comparison with Standard Biblatex

Standard biblatex	citemsp
<code>\autocite[§3.2]{smith2003}</code> → [1, §3.2]	<code>\citemsp{smith2003/3.2}</code> → [1 ^{§3.2}]
<code>\autocites[§3.2]{smith2003}</code> <code>[eq.~4.3]{meyerrochow2003}</code> → [1, §3.2][2, eq. 4.3]	<code>\citemsp{smith2003/3.2,</code> <code>meyerrochow2003/e4.3}</code> → [1 ^{§3.2} , 3 ^{eq.4.3}]
<code>\autocites[§3.2, ¶5]{smith2003}</code> → [1, §3.2, ¶5]	<code>\citemsp{smith2003/3.2/5}</code> → [1 ^{§3.2} ₅]
No built-in typed locator syntax; user must type symbols manually	Built-in prefixes; extensible via <code>\citemspprefix</code>

Table 2: Feature comparison between standard biblatex locators and citemsp.

6 Conventions

- **Section numbering:** use the source’s native format (e.g., 3.2.1, IV, A.1).
- **Paragraph counting:** count paragraphs from the start of the referenced section; as a manual convention.

Note: Paragraph locators rely on manual counting from the start of the referenced section. Since paragraphs are not explicitly numbered in published sources, these locators may be ambiguous across editions or between readers. When using ¶ locators, consider providing additional context in the surrounding text so the reader can locate the passage independently.

- **Subsection references:** use the full path (e.g., key/3.2.1).

References

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- [9] Te Faye Yap et al. “Necrobotics: Biotic Materials as Ready-to-Use Actuators”. In: *Advanced Science* 9 (2022), p. 2201174. DOI: [10.1002/advs.202201174](https://doi.org/10.1002/advs.202201174).